Discovering Genetics-related Biomarkers of Neuronal Vulnerability Using Post-mortem Datasets from Multiple Large Brain Banks

D-346

Xiaolin Zhou, M.S.c[1,2]; Daniel Kiss M.H.Sc[1,2];Isabel Kerrebijn[1,3]; Vanessa Gonçalves[1,2]; Yuliya Nikolova[1,2]; Shreejoy Tripathy[1,2]

1. Institute of Medical Sciences, University of Toronto, Toronto, ON, Canada; 2. Krembil Centre for Neuroinformatics, CAMH, Toronto, ON, Canada; 3. Lunenfeld-Tanenbaum Research Institute, Sinai Health, Toronto, ON, Canada

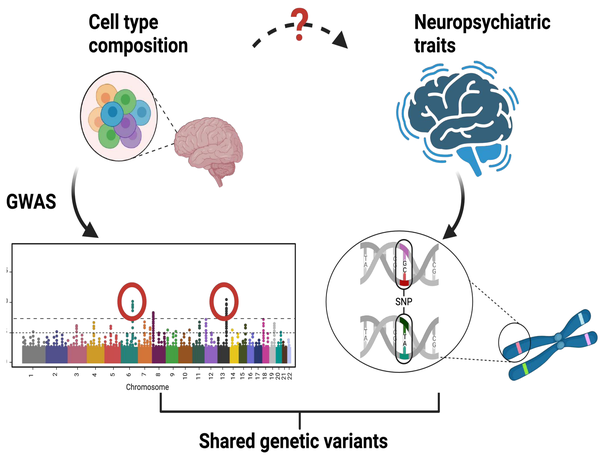
camh | Krembil Centre for Neuroinformatics

UNIVERSITY OF TORONTO

BACKGROUND

Objective:
Identify genetic factors governing brain cell-type vulnerabilities and their links to neuropsychiatric traits.

Hypothesis:
The selective vulnerability of different neuronal subtypes will be associated with distinct genetic risk factors.



METHODS

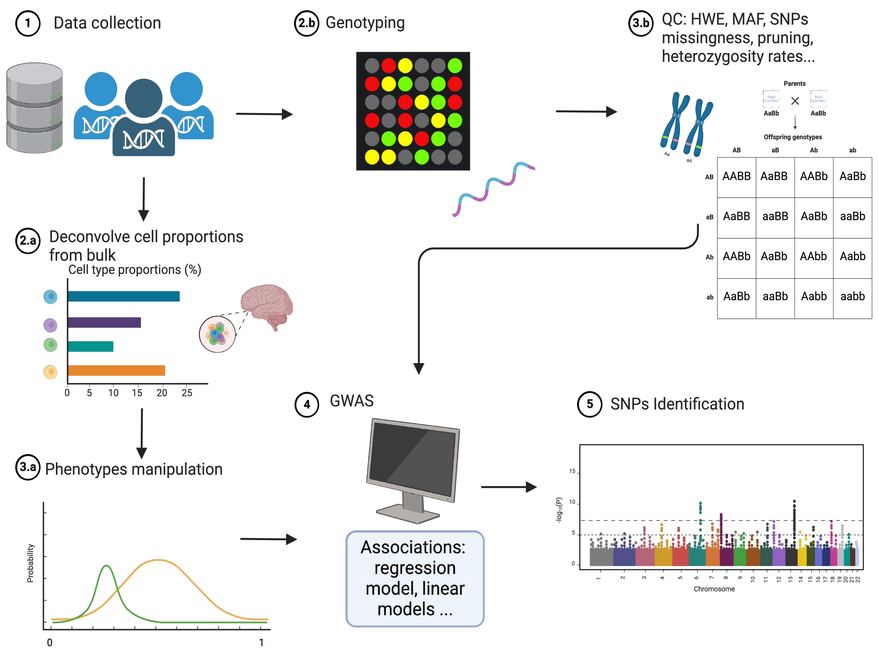


Figure 1. Schematic Overview of the Cell Type Deconvolution and GWAS analysis Workflow

Table 1. Demographic and Clinical Characteristics of datasets

Cohort / Dataset	N (EUR)	N (AFR)	Age (SD)	Sex (% M)	Brain region(s)	RNA type	DNA type	Cohort type
ROS/MAP	911	1	89.6 (6.6)	31	DLPFC, others	bulk / snRNAseq	WGS	aging/AD
MAYO	254	0	82.6 (8.0)	43	TCX	bulk RNAseq	WGS	aging/AD
MSBB	223	36	83.3 (7.4)	47	FCX, others	bulk RNAseq	WGS	aging/AD
CommonMind (MSBB, Penn, Pitt)	526	90	66.1 (18.1)	60	DLPFC	bulk / snRNAseq	SNP array	Psychiatry

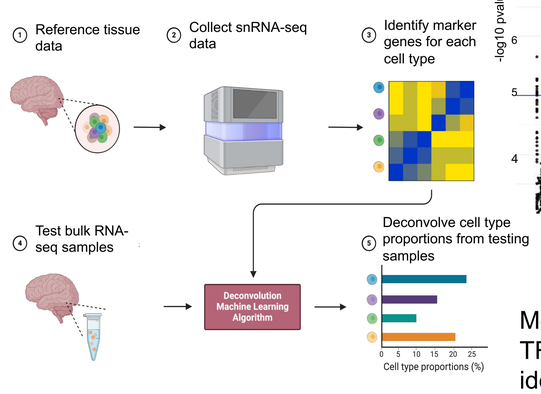
We benchmarked various cell-type deconvolution algorithms for bulk RNAseq data. Cell type proportion (CTP) estimates were validated against single-nucleus RNA sequencing (snRNA-seq) data from the same subjects.

Subsequently, we performed a genome-wide association study (GWAS) to discover genetic variants linked to changes in CTPs.

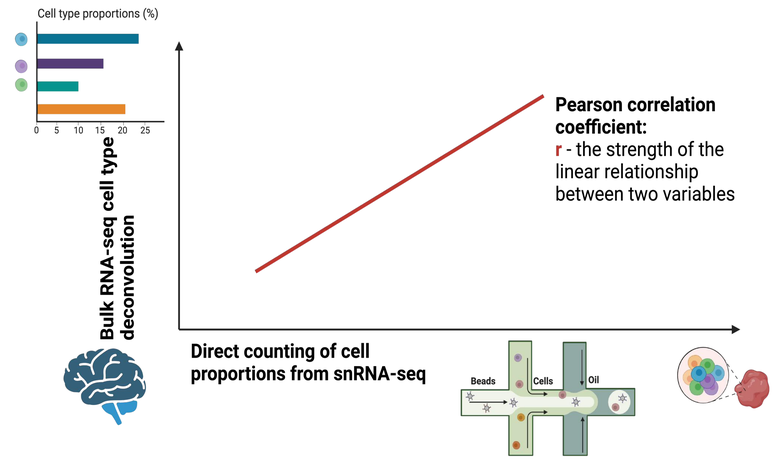
RESULTS

Deconvolving cell type proportions from bulk RNA–seq using Marker Gene Profiles (MGP)

- snRNAseq data used to generate cell-type specific marker lists
- MGP used to predict relative cell type proportions (rCTP) in bulk tissue from DLPFC
- rCTPs used in regression models to predict highly associated SNPs with cell type proportions



Benchmarking MGP-derived cell type estimations with snRNA-seq (ground truth)



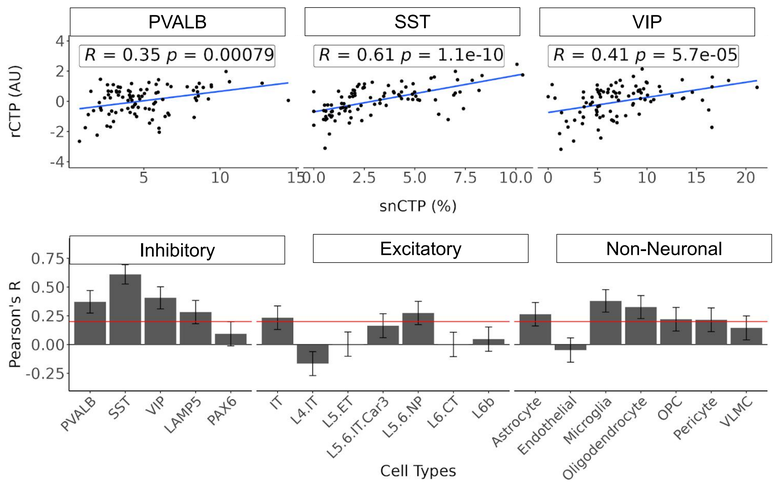
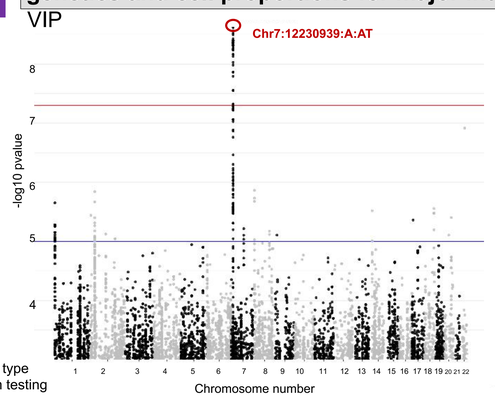


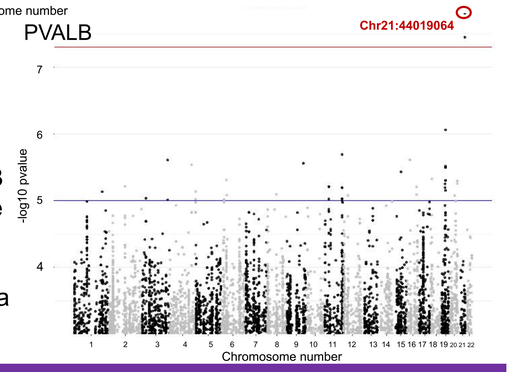
Figure 2. Correspondence between relative cell type proportions from bulk tissue deconvolution and single-nucleus RNAseq derived cell type proportions from matched tissue samples.

- For subjects in MSSM cohort with both bulk RNAseq and snRNAseq sampled from DLPFC, there is a moderate to relatively high degree of correspondence between relative cell type proportions (rCTPs) calculated using the Marker Genes Profile method and single-nucleus proportions.
- Across all cell types, we found that MGP-based rCTPs were usually positively correlated with snCTPs.
- With confidence in MGP-derived estimates, we used this cell type proportions as phenotype data for subsequent GWAS.

Preliminary GWAS results demonstrate associations between genetics and cell proportions for major interneurons



The SNP for VIP was found to be in the vicinity of the TMEM106B gene, which is a known modifier of neurodegenerative disease and cognitive aging.



Moreover, TRAPPC10 was identified near the top SNP for PVALB cells, and this gene is closely associated with neurodevelopmental disorder.

CONCLUSION

- In summary, our research demonstrates that brain cell type proportions can be reliably inferred from bulk brain tissue RNAseq datasets.
- Moreover, we find that these proportions are in part governed by genetics.

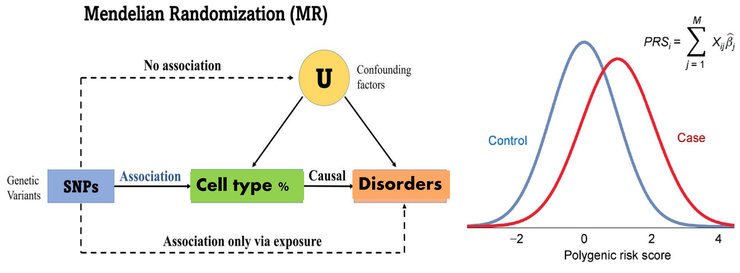
FUTURE DIRECTIONS

Increasing statistical power

- Apply the framework to more cohorts in AMP-AD consortium to obtain larger samples size for higher statistical power in finding cell type-specific SNPs

Post-GWAS analysis

- Use Mendelian Randomization to identify causal relationships between cell type proportions and neuropsychiatric disorders
- Calculate polygenic risk score (PRS) to provide a measure of brain disorders caused by genetic factors.



ACKNOWLEDGEMENT

This study was funded by CAMH, the Krembil Centre for Neuroinformatics, NeuroNex NSF-CIHR-DFG Team Grant, and NSERC Discovery grant.

Xiaolin is a Master of Medical Science student at the University of Toronto. She is in the TripLab lead by Dr. Shreejoy Tripathy. After accomplishing her Honors Bachelor of Science in Bioinformatics and Computational Biology at the University of Toronto, she is further pursuing her speciality in Neuroinformatics.

Dr. Shreejoy Tripathy is a scientist in the Krembil Centre for Neuroinformatics at the Centre for Addiction and Mental Health (CAMH). He is also an assistant professor in the Department of Psychiatry at the University of Toronto. His lab develops machine learning and statistical methods to help neuroscientists translate information at different levels of organization, from gene expression to neuron electrophysiology.

xiaolin.zhou@mail.utoronto.ca linkedin.com/in/xiaolin-cloris-zhou-1250a21b2/